Rockchip SD Card Boot Reference

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Rockchip Electronics Co., Ltd.

No.18 Building, A District, No.89, software Boulevard Fuzhou, Fujian, PRC

Website: www.rock-chips.com

Customer service Tel: +86-4007-700-590

Customer service Fax: +86-591-83951833

Customer service e-Mail: fae@rock-chips.com

Preface

Overview

This document primarily introduces various uses of SD cards by Rockchip, including Firmware creation, generation of SD cards with different functionalities, distribution of Firmware within the SD card, and the boot process. Engineers can follow this document to troubleshoot various issues encountered during SD card usage.

Intended Audience

This document (this guide) is primarily intended for the following engineers:

Technical Support Engineers

Software Development Engineers

Revision History

Version	Author	Date	Description
V1.0.0	Zhu Zhizhan, Liu Yi	2018-07-17	Initial version
V1.1.0	Zhu Zhizhan	2019-09-06	Added GPT support for upgrade cards
V1.2.0	Zhu Zhizhan	2021-04-15	Added BCB flag description and various sections
V1.3.0	Zhu Zhizhan	2022-01-12	Added RK3588 support
V1.3.1	Zhu Zhizhan, Huang Tao	2025-01-07	Changed document classification level. Corrected document format.

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1. Type Overview

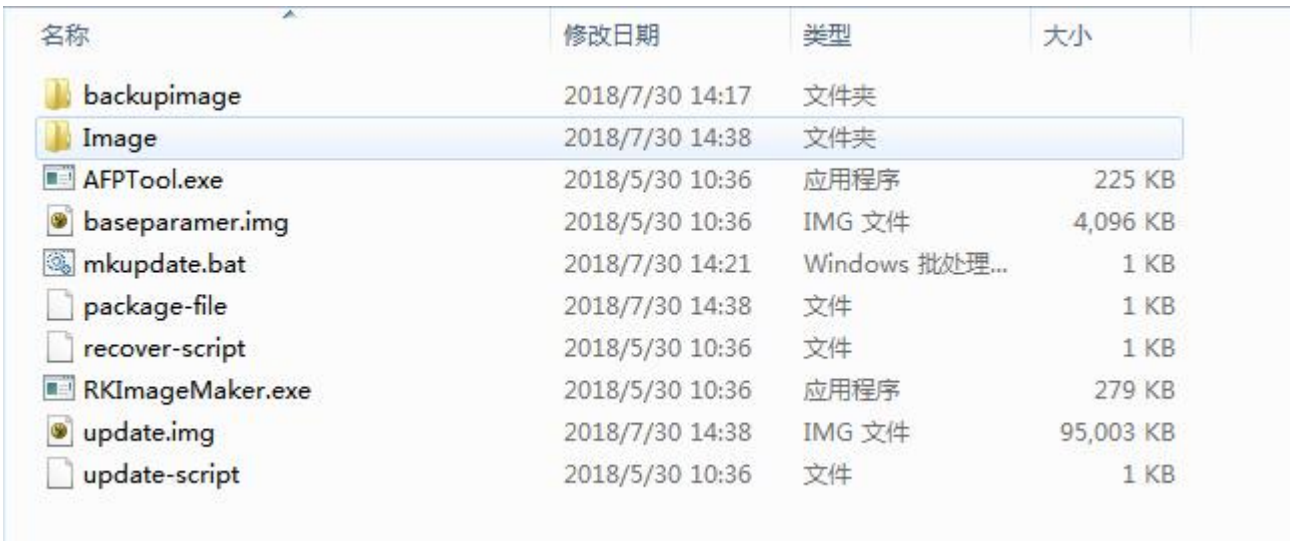
Rockchip classifies SD cards into standard SD cards, SD upgrade cards, SD boot cards, and SD Repair cards. Different card types can be created by downloading the `update.img` to an SD card using the SDDiskTool.

Card Type	Function
Standard SD Card	A standard storage device
SD Upgrade Card	The device boots from the SD card into recovery mode, where the recovery process is responsible for updating the device storage with the Firmware from the SD card
SD Boot Card	The device boots directly from the SD card
SD Repair Card	Copies the Firmware from the SD card to the device storage starting from the pre-loader

2. Firmware Creation

The `update.img` is a collection provided by Rockchip containing the complete Firmware suite. It includes not only the full Firmware but also data such as Firmware integrity checks. The `update.img` enables users to conveniently update the entire Firmware suite.

Rockchip provides dedicated tools for creating `update.img`. If using the Rockchip SDK, navigate to the `RKTools/linux/Linux_Pack_Firmware/rockdev/` directory as follows:



名称	修改日期	类型	大小
backupimage	2018/7/30 14:17	文件夹	
Image	2018/7/30 14:38	文件夹	
AFPTool.exe	2018/5/30 10:36	应用程序	225 KB
baseparamer.img	2018/5/30 10:36	IMG 文件	4,096 KB
mkupdate.bat	2018/7/30 14:21	Windows 批处理...	1 KB
package-file	2018/7/30 14:38	文件	1 KB
recover-script	2018/5/30 10:36	文件	1 KB
RKImageMaker.exe	2018/5/30 10:36	应用程序	279 KB
update.img	2018/7/30 14:38	IMG 文件	95,003 KB
update-script	2018/5/30 10:36	文件	1 KB

The required update.img can be packaged by modifying the package-file. The contents of the package-file are as follows:

```
# NAME     Relative path
#
#HWDEF     HWDEF
package-file    package-file
bootloader  Image/MiniLoaderAll.bin
parameter   Image/parameter.txt
uboot       Image/uboot.img
trust       Image/trust.img
misc        Image/misc.img
resource    Image/resource.img
kernel      Image/kernel.img
boot        Image/boot.img
#recovery   Image/recovery.img
#system     Image/system.img
#vendor     Image/vendor.img
#oem        Image/oem.img
# baseparamer   Image/baseparamer.img
# The file to be written to the backup partition is the image itself (update.img)
# SELF is a keyword indicating the upgrade file (update.img) itself.
# Do not include the content of the SELF file when generating the upgrade file, but record
it in the header information.
# Do not unpack the contents of the SELF file when unpacking the upgrade file.
# Do not package RESERVED as backup
backup     backupimage/backup.img
update-script   update-script
recover-script  recover-script
```

When adding files, write the filename and firmware address. If a certain firmware does not need to be packaged, prefix the firmware name with "#" to disable it.

Click to run mkupdate.sh to generate update.img.

3. Tool Usage

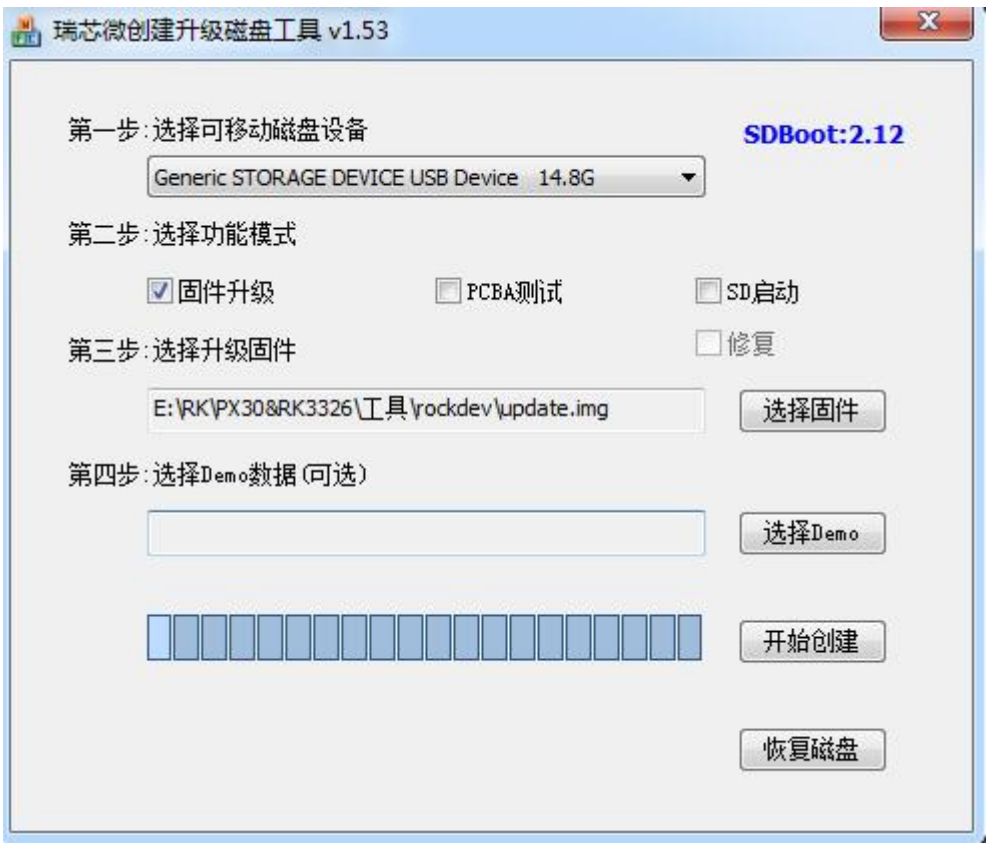
3.1 Standard SD Card

A standard SD card is used exactly like on a PC and can be utilized as ordinary storage space within the U-Boot and Kernel systems. No tools are required to perform any operations on the SD card.

3.2 SD Upgrade Card

An SD upgrade card is created using Rockchip tools to enable system upgrades for local storage (such as eMMC or NAND flash) via the SD card. SD card upgrading is a firmware upgrade method that operates independently of a PC or network. Specifically, the SD card boot code is written to the reserved area of the SD card, and the Firmware is copied to the visible partition of the SD card. When the main controller boots from the SD card, the SD card boot code and upgrade code flash the Firmware into the local main storage. Additionally, the SD upgrade card supports PCBA testing and the copying of Demo files. These features of the SD upgrade card allow firmware upgrades to be performed without a PC, thereby improving production efficiency.

The procedure for creating an SD upgrade card is as follows:



Proceed as follows:

1. Select the removable disk device.
2. Set the function mode to Firmware Upgrade.
3. Select the firmware to be upgraded.

4. Click Start Creation.

Refer to the figure above for specific configuration settings.

Re-creation:

If the SD upgrade card has already been created and only the Firmware and demo files need to be updated, proceed as follows:

1. Copy the Firmware to the root directory of the SD card and rename it to sdupdate.img.
2. Copy the demo files to the Demo directory under the root directory of the SD card.

SD Upgrade Card Format (Non-GPT)

Offset	Data Segment
Sector 0	MBR
Sectors 64-4M	IDBLOCK (Boot flag set to 0)
4M-8M	Parameter
8M-12M	uboot
12M-16M	trust
.....	misc (BCB writes recovery\n--rk_fwupdate\n)
.....	resource
.....	kernel
.....	recovery
Remaining Space	FAT32 stores update.img

SD Upgrade Card Format (GPT)

Offset	Data Segment
Sector 0	MBR
Sectors 1-34	GPT Partition Table
Sectors 64-4M	IDBLOCK (Boot flag set to 0)
4M-8M	Parameter
.....	uboot
.....	trust
.....	misc
.....	resource
.....	kernel
.....	recovery
Remaining Space	FAT32 stores update.img

3.3 SD Boot Card

The SD boot card is created using Rockchip tools to enable the device system to boot directly from the SD card. This greatly facilitates users in updating newly compiled Firmware without flashing it to the device storage, and it can also serve as the device's primary storage. It is currently mainly used for system booting from an SD card or for PCBA testing. **Note**: PCBA testing is merely a function item under recovery and can be used for both upgrade cards and boot cards.

The process for creating a boot card is as follows:



1. Select the removable disk device.
2. Set the function mode to SD Boot.
3. Select the firmware to be upgraded.
4. Click Start to create.

Refer to the figure above for specific configuration settings.

SD Boot Card Format (Non-GPT)

Offset	Data Segment
Sector 0	MBR
Sectors 64-4M	IDBLOCK (Boot flag set to 1)
4M-8M	Parameter
8M-12M	uboot
12M-16M	trust
.....	misc
.....	resource
.....	boot
.....	kernel
.....	recovery
.....	system
.....	user

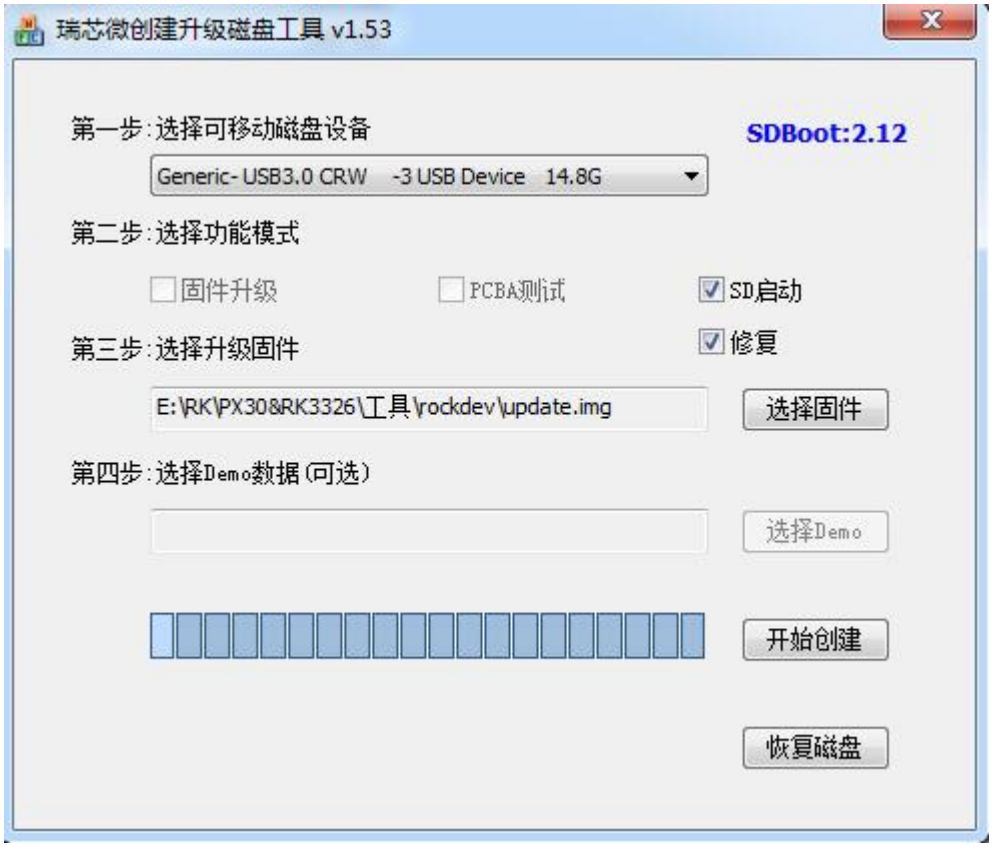
SD Boot Card Format (GPT)

Offset	Data Segment
Sector 0	MBR
Sectors 1-34	GPT Partition Table
Sectors 64-4M	IDBLOCK (Boot flag set to 1)
.....	uboot
.....	boot
.....	trust
.....	resource
.....	kernel
.....	recovery
.....	system
.....	vendor
.....	oem
.....	user
Last 33 sectors	Backup GPT

3.4 SD Repair Card

The SD Repair Card operates similarly to the SD card upgrade function, but the firmware upgrade occurs within the SD card upgrade code of the pre-loader (miniloader). First, the tool writes the boot code to the reserved area of the SD card, then copies the Firmware to the visible partition of the SD card. When the SoC boots from the SD card, the SD card upgrade code upgrades the Firmware to the local primary storage. This method is primarily used to recover devices when the device Firmware is corrupted.

The procedure for creating a recovery card is as follows:



1. Select the removable disk device.
2. Set the function mode to SD Boot and Repair.
3. Select the firmware to be upgraded.
4. Click Start to create.

Refer to the figure above for specific configuration settings.

SD Repair Card Format (Non-GPT)

Offset	Data Segment
Sector 0	MBR
Sectors 64-4M	IDBLOCK (Boot flag set to 2)
4M-8M	Parameter
8M-12M	uboot
12M-16M	trust
.....	misc
.....	resource
.....	boot
.....	kernel
.....	recovery
.....	system
.....	user

SD Repair Card Format (GPT)

Offset	Data Segment
Sector 0	MBR
Sectors 1-34	GPT Partition Table
Sectors 64-4M	IDBLOCK (Boot flag set to 2)
.....	uboot
.....	boot
.....	trust
.....	resource
.....	kernel
.....	recovery
.....	system
.....	vendor
.....	oem
.....	user
Last 33 sectors	Backup GPT

4. Flag Description

SD cards serving various functions include specific flags within the card.

At sector 64 of the SD card, if the starting magic number is 0xFCDC8C3B, the card is identified as a special card, prompting the device to read Firmware from the SD card for booting. Otherwise, it is treated as a standard SD card. At the location (sector 64 + 616 bytes), various card flags are stored. Currently, there are three types:

Flag	Card Type
0	Upgrade Card or PCBA Test Card
1	Boot Card
2	Repair Card

Currently, this method of writing the idb block flag has the following disadvantages:

- Proprietary RK design, lacking compatibility

- Security risks, as this flag is written to the idb block during card manufacturing, compromising the integrity of the idb block
- The new version of the idb block does not reserve space for this flag (which is unnecessary, as it would encounter the aforementioned issues)

To address these issues, RK reuses the Android BCB design by adding `recovery\n--rk_fwupdate\n` in recovery as the flag for entering SD upgrade mode.

Support status for both flag methods:

Platform	idb block flag	Android BCB
rk3588		✓
rk3568/rk3566		✓
rv1126/rv1109	✓	✓
rk3399	✓	✓
rk3368	✓	✓
rk3328	✓	✓
rk3326	✓	✓
rk3308	✓	✓
rk3288	✓	✓
rk3229	✓	✓
rk3128	✓	✓
rk3126	✓	✓

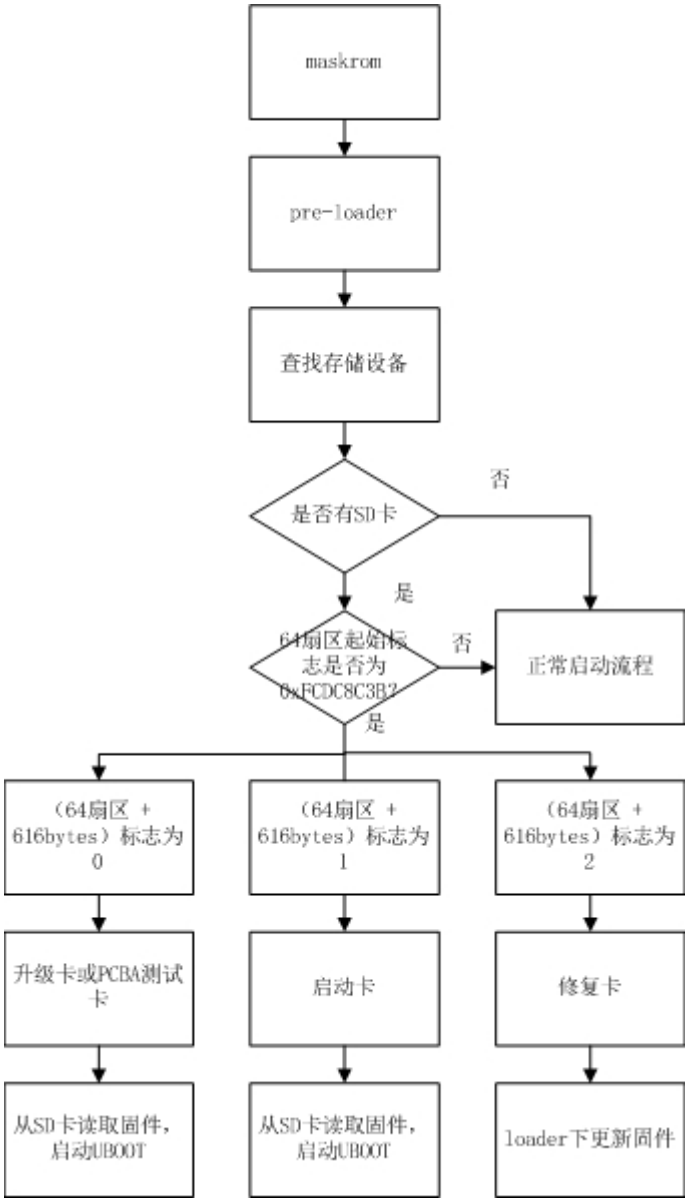
With future platform upgrades and tool updates, the method of writing the IDB block flag will be gradually phased out.

Note: SDDiskTool must be updated to v1.67 or later to support the Android BCB method.

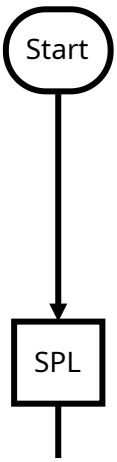
5. Process Analysis

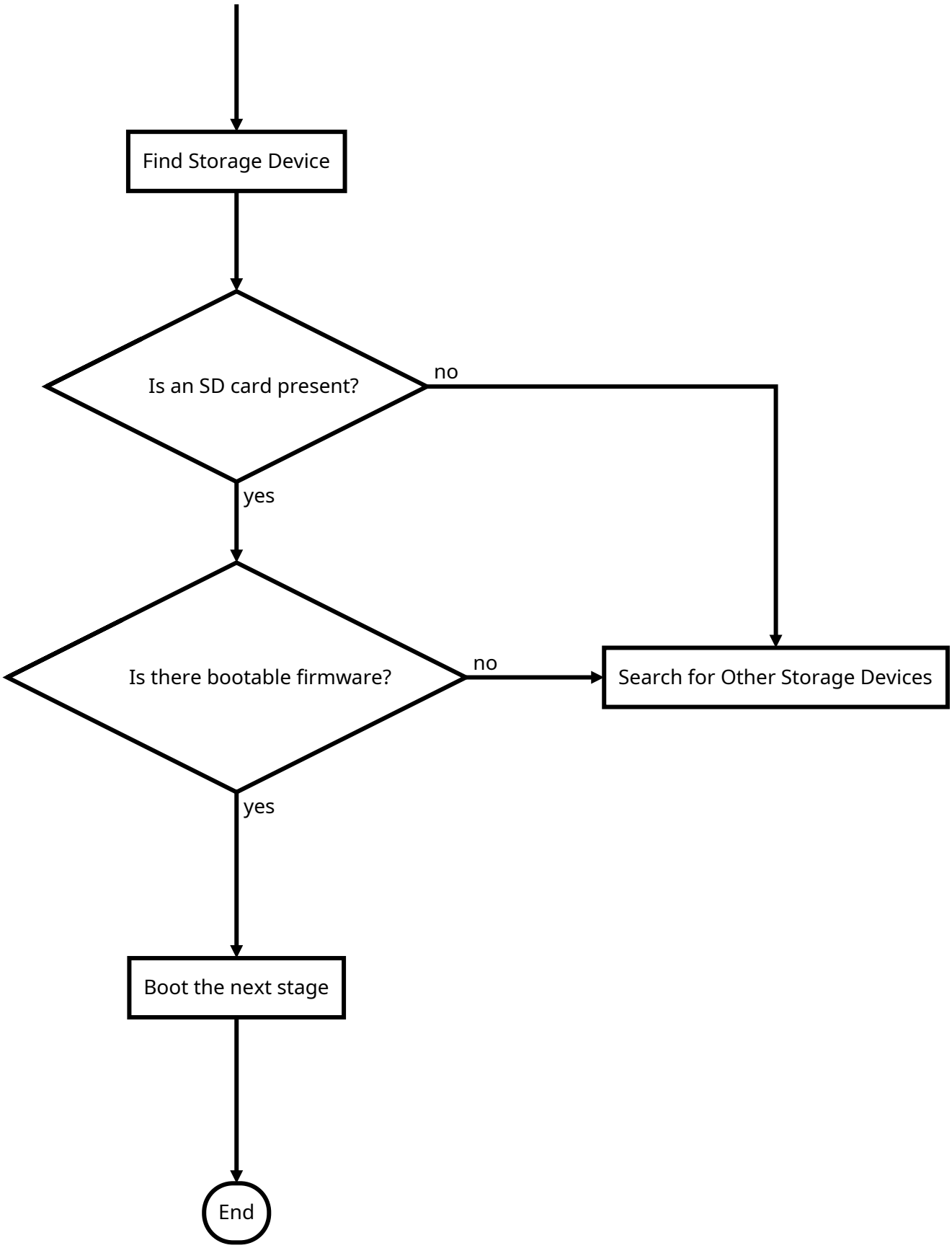
The SD card boot process consists of the pre-loader/SPL boot sequence and the U-Boot boot sequence. Both the pre-loader and U-Boot sequences require loading and detecting the SD card as well as the Startup Flag within the SD card's IDB Block, executing different functions based on these flags. The SPL sequence sets the SD card as the highest-priority boot device; if bootable Firmware exists on the SD card, it is loaded and booted from the SD card with priority.

5.1 pre-loader Boot Sequence



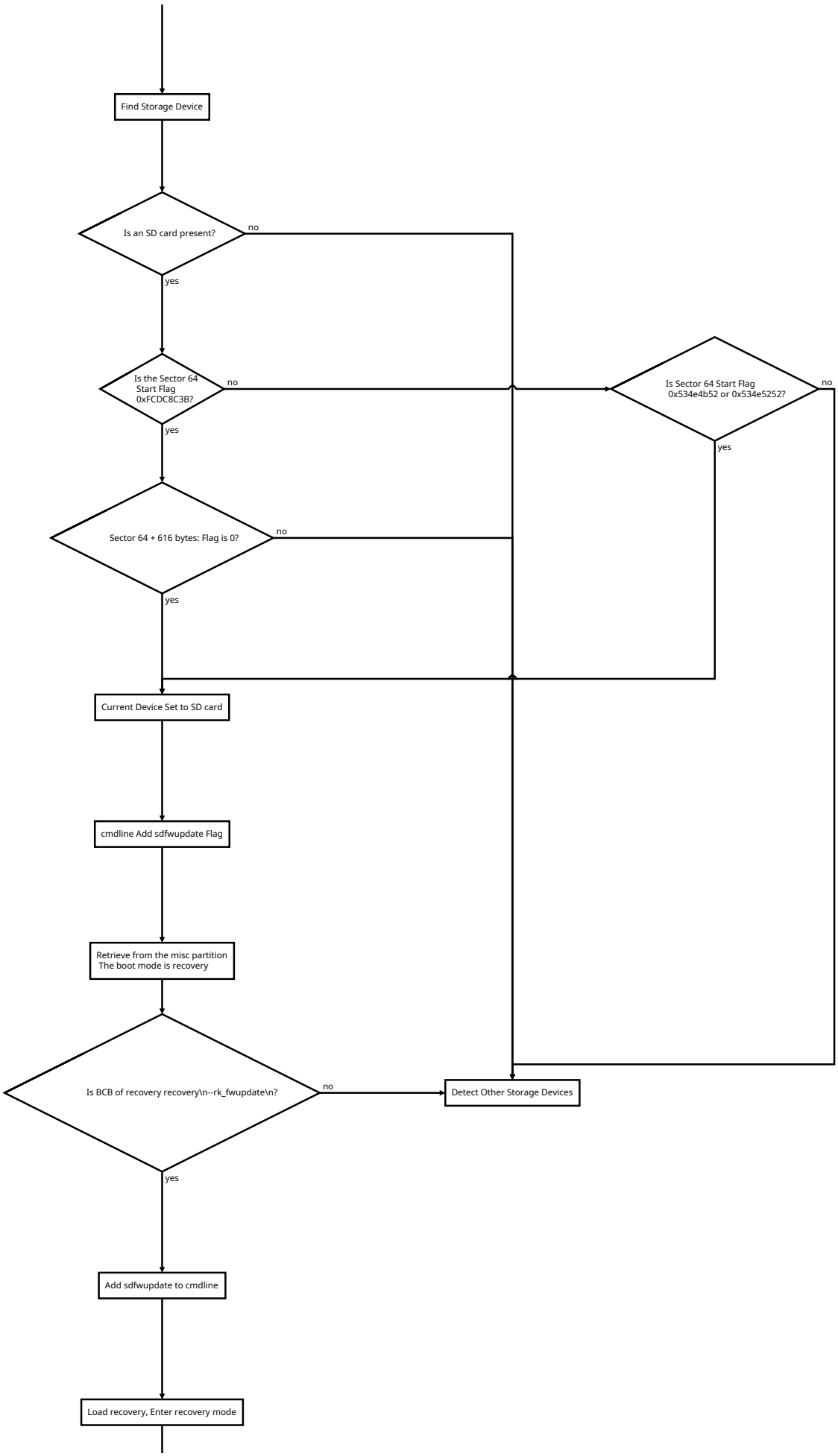
5.2 SPL Boot Flow





5.3 U-Boot Boot Flow


Start
Uboot


Find Storage Device
Is an SD card present?
no
yes
Is the Sector 64 Start Flag 0xFCDC8C3B?
no
yes
Is Sector 64 Start Flag 0x534e4b52 or 0x534e5252?
no
yes
Sector 64 + 616 bytes: Flag is 0?
no
yes
Current Device Set to SD card
cmdline Add sdfwupdate Flag
Retrieve from the misc partition The boot mode is recovery
Is BCB of recovery recovery\n--rk_fwupdate\n?
no
Detect Other Storage Devices
yes
Add sdfwupdate to cmdline
Load recovery, Enter recovery mode



5.4 Recovery and PCBA Instructions

For details, refer to "Rockchip Recovery User Guide V1.03.pdf" (available in the Rockchip SDK documentation package under the recovery guide directory).

6. Precautions

1. For non-GPT formats, U-Boot must be configured with CONFIG_RKPARM_PARTITION.
2. When creating an SD upgrade card, update.img must include the following firmware: MiniLoaderAll.bin, parameter.txt, uboot.img, trust.img, misc.img, resource.img, and recovery.img. Otherwise, flashing update.img will result in an MBR write failure error.